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AI Health System for Doctors and Patients

Project Constraints and Impacts Document

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1. Summary of Compliance with Standards (Applied Engineering Standards)

This section summarizes the engineering and regulatory standards considered during the project. For each standard, the compliance level is reported as Compliant, Partially Compliant, or Not Applicable, based on available design documents, implementation evidence, test artifacts, and project scope limitations.

No.	Standard	Status
1	**IEEE 830-1998 (Software Requirements Specification)**	Compliant
2	**IEEE 1016-2009 (Software Design Description)**	Compliant
3	**UML 2.x Modeling Principles**	Compliant
4	**ISO/IEC 25010 (Software Quality Model)**	Compliant
5	**PEP 8 (Python Coding Style Guidelines)**	Compliant
6	**GDPR (General Data Protection Regulation)**	Compliant
7	**KVKK (Turkish Personal Data Protection Law)**	Partially Compliant
8	**HIPAA (Health Insurance Portability and Accountability Act)**	Not Applicable

1.1. IEEE 830-1998 (Software Requirements Specification)

Justification: The project requirements, constraints, and major functional and non-functional expectations were documented in a structured way.
Evidence References: Project Specifications Report, Analysis Report, Test Plan Report, Requirement Traceability Matrix.

1.2. IEEE 1016-2009 (Software Design Description)

Justification: The project produced both high-level and low-level design documents describing the architecture, subsystem decomposition, interfaces, workflows, persistence strategy, error handling, and testability approach.
Evidence References: *High-Level Design Report*, *Low-Level Design Report*.

1.3. UML 2.x Modeling Principles

Justification: UML was used extensively to document the system structure and behavior, including class diagrams, sequence diagrams, activity diagrams, and use case diagrams.
Evidence References: *Low-Level Design Report*, UML sections.

1.4. ISO/IEC 25010 (Software Quality Model)

Justification: The project explicitly considered quality attributes such as security, maintainability, usability, reliability, and performance.
Evidence References: *High-Level Design Report*, *Low-Level Design Report*, *Test Plan Report*.

1.5. PEP 8 (Python Coding Style Guidelines)

Justification: The backend implementation generally followed readable and modular Python coding practices, including clear naming and layered organization.
Evidence References: Backend implementation, *Low-Level Design Report*.

1.6. GDPR (General Data Protection Regulation)

Justification: Privacy-aware design principles were incorporated into the project through consent handling, role-based access control, audit logging, and restricted patient-facing output behavior.
Evidence References: *Project Specifications Report*, *Analysis Report*, backend consent and authorization workflows.

1.7. KVKK (Turkish Personal Data Protection Law)

Justification: The system design reflects KVKK-related concerns through consent enforcement, restricted access to sensitive data, and traceability mechanisms. Nevertheless, full formal legal compliance verification was outside the scope of the project.
Evidence References: *Project Specifications Report*, *Analysis Report*, backend access-control and consent mechanisms.

1.8. HIPAA (Health Insurance Portability and Accountability Act)

Justification: HIPAA was considered only as a reference framework for privacy-sensitive healthcare software design. Since the system was not deployed as a certified production healthcare product under HIPAA processes, this standard was treated as not applicable in a formal compliance sense.
Evidence References: *Project Specifications Report*, *Analysis Report*.

2. Summary of Constraints Management

The project was developed under a set of practical constraints related to time, computational resources, data availability, deployment maturity, legal scope, and validation capacity. During implementation, these constraints were managed not by removing the project's core objectives, but by prioritizing a controlled and modular system design. Instead of attempting to build a clinically deployable end product, the project focused on delivering a technically functional, traceable, and safety-aware decision-support platform within the limits defined in the PKE-Plan.

One of the main constraints was computational and infrastructure capacity. The project involved training and integrating multiple medical imaging models across different tasks and datasets, which created significant pressure on available GPU time, storage, and experimentation cycles. To manage this, the team adopted a task-specific model selection strategy rather than attempting to optimize every architecture equally for every task. Only the best-performing models for selected tasks were integrated into the final platform, while other experiments remained part of the model-comparison process. This helped preserve the project's multimodal scope while keeping implementation feasible within available resources.

A second major constraint was the availability and suitability of datasets. Although multiple public datasets were used, they differed in size, class balance, annotation quality, and structure. These differences affected both training stability and the comparability of results across tasks. To address this, the project used task-specific

preprocessing, split management, and model selection decisions rather than forcing all datasets into a single unified pipeline. Nevertheless, dataset limitations also led to scope boundaries. The system could demonstrate strong prototype-level functionality, but not claim broad real-world generalization or clinical robustness across all populations and settings.

Time and project-scope limitations also affected the final feature set. The initial vision of the project included broader experimentation and more advanced model outputs in some areas. However, given the limited semester timeline, the implementation prioritized features that directly supported the platform's main value proposition: doctor-side case review, patient-side symptom interaction, safe AI response control, auditability, and multi-model image analysis. As a result, some advanced experimental directions were not fully integrated into the deployed platform. For example, segmentation-oriented work such as the brain tumor U-Net study remained an experimental extension rather than a production-integrated workflow. This was a deliberate scope decision made to preserve stability, coherence, and reportable completion of the core system.

Another important constraint concerned external dependencies. The symptom-analysis pipeline relied on an external HPC-hosted language model service, which introduced risks related to latency, availability, and response variability. Instead of exposing this dependency directly to end users, the project addressed the risk through application-level control mechanisms such as consent verification, moderation, disclaimer attachment, fallback handling, and role-aware response shaping. This design decision reduced the operational risk caused by the external service while allowing the system to preserve its patient-support functionality.

Legal and regulatory scope was also managed carefully. The project considered privacy-sensitive healthcare design principles and aligned itself conceptually with standards and regulations such as GDPR and KVKK. However, full legal certification, institutional compliance verification, and production-grade regulatory validation were outside the scope of the project. This constraint was handled by treating privacy, access control, and auditability as engineering priorities, while clearly limiting any claim of formal legal or clinical compliance.

These constraints resulted in several controlled deviations from an ideal full-scale healthcare AI deployment. The final system should therefore be understood as an advanced engineering prototype rather than a clinically validated medical product. Even so, the project successfully achieved its most important planned outcomes: integration of multiple AI models, implementation of role-specific workflows, enforcement of safety-oriented controls, and creation of a modular and traceable architecture.

The main lesson learned is that constraints in healthcare-related AI projects are not merely obstacles; they actively shape better engineering decisions. Resource limits encouraged selective integration, legal limits encouraged clearer safety boundaries, and

deployment limits reinforced the importance of modularity and human oversight. Overall, the project showed that responsible scope control is itself an engineering achievement, especially in sensitive domains such as AI-assisted healthcare.

3. Expected and Realized Impact Results

This section presents the expected and realized impacts of the AI Health System for Doctors and Patients across multiple dimensions. Due to the scope of the project and the absence of large-scale real-world deployment, impacts are evaluated primarily through system capabilities, implemented functionalities, and qualitative observations. Nevertheless, the system demonstrates a well-structured and feature-complete prototype integrating multiple AI models, datasets, and controlled workflows.

3.1. Economic Impact

Expected Impact:
The system was expected to improve efficiency in healthcare workflows by assisting early-stage evaluation and reducing repetitive tasks.

Realized Impact:
The implemented platform integrates multiple automated workflows, including medical image analysis and symptom-based interaction, which reduce the need for manual preprocessing and preliminary evaluation. The system supports multiple diagnostic tasks across different datasets, indicating potential efficiency gains in clinical environments. Although no quantitative cost analysis was conducted, the integration of AI-based automation demonstrates a foundation for reducing operational workload.

3.2. Social Impact

Expected Impact:
The project aimed to improve access to preliminary medical guidance through a digital and user-friendly platform.

Realized Impact:
The system provides a role-aware interface supporting both doctors and patients. Patients can submit symptoms through a structured interaction pipeline and receive simplified, controlled responses. Doctors are provided with more detailed outputs and control mechanisms such as annotation and override. This separation of roles demonstrates the system's ability to provide accessible and context-aware interaction, improving usability and reducing potential misinterpretation.

3.3. Environmental Impact

Expected Impact:
The system was expected to have minimal environmental impact, with potential indirect benefits through reduced physical interactions.

Realized Impact:
The system operates entirely as a digital platform, supporting remote interaction and reducing reliance on physical resources in early-stage evaluation processes. While no direct environmental measurements were conducted, the architecture supports a shift toward more resource-efficient and remote-access healthcare solutions.

3.4. Legal Impact

Expected Impact:
The system was expected to consider data protection and privacy principles aligned with regulations such as GDPR and KVKK.

Realized Impact:
The system incorporates privacy-aware design elements, including authenticated access, controlled exposure of outputs, and structured audit logging. These mechanisms demonstrate alignment with fundamental data protection principles. However, formal legal compliance verification and certification were not performed within the scope of the project.

3.5. Health Impact

Expected Impact:
The system aimed to support healthcare processes by providing AI-assisted insights for medical decision support.

Realized Impact:
The system integrates multiple deep learning models, including ResNet50, DenseNet121, EfficientNet-B4, and Vision Transformer (ViT), trained on diverse medical datasets such as NIH Chest X-ray, BraTS brain tumor datasets, COVID radiography, breast ultrasound, fracture detection, and multi-region X-ray datasets.

These models support multiple diagnostic tasks across different imaging modalities within a unified platform. In addition, a symptom-based interaction pipeline is integrated through an external language model service.

Doctors can review AI-generated outputs, provide annotations, and override results, ensuring human control over decision-making. Patients receive simplified and moderated outputs with disclaimers. While no clinical validation was performed, the system demonstrates a functional multi-model, multi-modal decision-support capability.

3.6. Security Impact

Expected Impact:
The system was expected to ensure secure handling of sensitive medical data through authentication, authorization, and monitoring mechanisms.

Realized Impact:
The system implements a structured authentication mechanism including login, registration, and logout workflows. Protected endpoints are enforced through user-based access control mechanisms.

In addition, a dedicated audit logging service records system-level events, including user actions, entity types, operation status, and IP addresses, enabling strong traceability.

Furthermore, system monitoring capabilities are implemented through a health-check service that tracks resource usage such as CPU and memory.

These components collectively demonstrate a security-aware, traceable, and observable system design. However, comprehensive security validation such as penetration testing was beyond the scope of the project.